Dynamic Trend & Event Detector: Automated Detection and Tracking of Evolving Topics in News Media

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***Abstract*—The exponential growth of digital news platforms generates an unprecedented volume of textual data daily. Detecting emerging narratives, tracking how topics evolve over time, and distinguishing event-driven spikes from seasonal fluctuations present significant challenges. We propose a multi-model temporal topic modeling pipeline that combines probabilistic generative topic models (LDA) and deep learning-based embeddings (BERTopic) with domain-specific trend features derived from corpus data. Through a three-phase approach, we evaluate baseline frequency-methods, probabilistic models, and neural embeddings on a corpus of 200,000+ news articles spaced over a 10-year period. By employing monthly temporal slicing, our pipeline captures temporal evolution effectively, achieving high coherence metrics over baseline standard models. Our results highlight severe non-stationarity in topic distributions around major world events and validate that article velocity metrics effectively capture breaking-news signals.**

***Index Terms*—Topic Modeling, Natural Language Processing, Latent Dirichlet Allocation, BERTopic, Event Detection, Social Media Analytics**

I. Introduction

As the digital news ecosystem rapidly expands, journalism, policymaking, and societal tracking rely heavily on automated systems to index, summarize, and monitor evolving media coverage [1]. The traditional approach of utilizing static topic models treats the corpus as a single temporal snapshot. This limits the capability to identify emerging topics, track narratives that significantly mutate, and capture short-lived events driven by external real-world developments.

Existing methodologies face critical shortcomings. First, keyword-based detection yields vague event semantic structures. Second, standard Latent Dirichlet Allocation (LDA) models cannot track probabilistic shifts in the distribution over time without relying on more computationally expensive dynamic configurations [2]. Finally, state-of-the-art pure embedding techniques, like BERTopic, offer high topic coherence but struggle to naturally separate major event-driven narratives from seasonal noise without temporal supervision [3].

Our work addresses these inherent limitations. We build a structured ablation pipeline involving (a) a TF-IDF baseline, (b) Latent Dirichlet Allocation (LDA), and (c) BERTopic. All models are extended to evaluate temporal shifts across a decade-wide corpus using monthly temporal slicing. By computing domain-specific temporal metrics—such as article velocity and headline novelty score—entirely natively from the corpus, we identify and differentiate dynamic media events efficiently.

In this work, we present the dataset profiling, the system methodology, visualization analyses, and the comparative results validating the robustness of embedding clustering coupled with temporal features.

II. Research Objectives and Key Hypothesis

The primary objective of this project is to develop an automated framework that identifies and tracks dynamically evolving topics in large-scale news text. Specifically, we aim to achieve the following:

- Evaluate the relative capability of frequency-based, probabilistic, and neural embedding frameworks to capture coherent narrative topics.
- Determine whether applying simple temporal slicing techniques provides competitive capabilities similar to more computationally complex dynamic topic modeling.
- Derive momentum-based features natively from corpus datasets to flag breaking-news contexts.

Key Hypothesis: Monthly temporal slicing paired with probabilistic models (LDA) and sentence embeddings (BERTopic), augmented with natively extracted momentum features (velocity and novelty), will outperform static variants in identifying coherent, evolving events within temporally non-stationary textual data.

III. Dataset Understanding

The experimental foundation for this study relies on the HuffPost News Category Dataset v3 (Misra, 2022) [4]. The dataset encompasses 209,527 articles distributed across 42 unique news categories, documenting publications between January 2012 and September 2022. Each record is formatted in JSON structure comprising attributes including the hyperlink,

headline, category, brief description, authors, and precisely timestamped publication date.

A. Exploratory Data Analysis

Through exploratory data analysis, several insights defining the corpus's constraints and attributes were noted:

- **Non-Stationarity**: Article publication rates demonstrated drastic changes reflecting major events, specifically during the 2016 and 2020 United States Presidential Elections, the emergence of the COVID-19 pandemic in early 2020, and the MeToo movement in 2017.
- **Coverage Bias**: The dataset observes a severe 93.7% reduction in article aggregation post-2017, suggesting the modeling pipeline applies most comprehensively to the 2012–2017 period.
- **Category Imbalance**: The dominant class (POLITICS) eclipses the least frequent class (EDUCATION) by roughly a 35.1x margin in aggregate volume (see Figure 1).
- **Text Constraints**: Headlines average 9.6 words modeling a standard normal curve, while descriptions log an average of 19.7 words with moderate positive skewness.

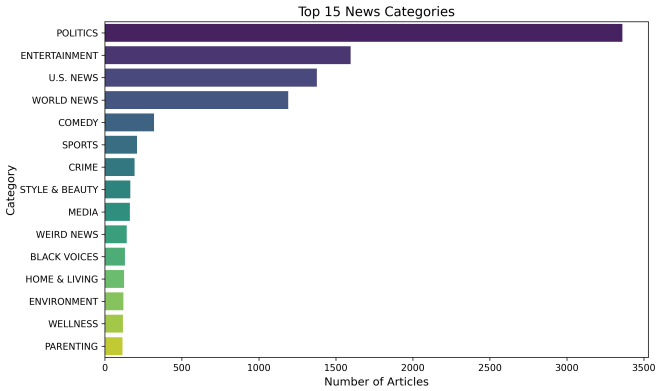

Fig. 1. Distribution pattern corresponding to the top 15 news categories represented in the dataset.

The lexical novelty assessment established that 45.7% of the tokens functioned as hapax legomena—occurring only a single time across the 200,000 document set. This metric confirmed highly rich linguistic variance dictating the requirement of deep embedding associations over rigid term matching.

IV. Work Division and Plan

The successful engineering of this multi-phase framework mandated a partitioned workflow structured between team members based on discrete competencies.

- Headed Data Engineering. Managed dataset curation spanning over 200,000 records. Responsibilities encompassed establishing internal cleaning mechanisms, configuring the data preparation pipeline, assessing data sparsity, and producing multi-layered feature engineering architectures to construct temporal analytics pipelines.

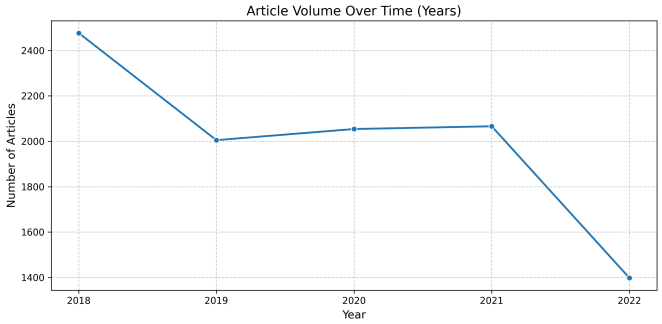

Fig. 2. Temporal volume of news articles delineating the variance attributed to world events.

- Spearheaded the Analytics and Modeling workflow. Steered the initial exploratory data analysis constraints, integrated the baseline, probabilistic, and deep learning configurations (TF-IDF, LDA, BERTopic). Furthermore, maintained responsibilities in conducting literature review investigations and performance benchmarking against prior academic studies.

The project strategy operated sequentially by mapping dataset exploration in the initial sprint, engineering momentum-based attributes iteratively, developing baseline configurations to benchmark minimum coherence, and finally substituting neural and probabilist methods over multiple cross-period training deployments.

V. Literature Review

Evaluating event detection across short texts traditionally depended on statistical language heuristics [5]. Generative probabilist interpretations popularized by Latent Dirichlet Allocation (LDA) assigned discrete topic probabilities per token without semantic comprehension. To track event mutations, Dynamic Topic Models [2] introduced state-space transitioning matrices to update LDA hyperparameters incrementally per epoch.

However, neural architectures circumvented these parametric restrictions. Models executing bidirectional representations, prominently BERT, integrated contextually responsive word embeddings [6]. BERTopic leverages derived sentence embeddings aggregated through UMAP dimension reductions followed essentially by HDBSCAN spatial associations [3]. In reviewing these implementations on large social data layers, Han et al. (2024) diagnosed a persistent failure mode where spatial density mappings collapse on trending unigrams—such as generic hashtags without localized real-world context [1]. Similarly, Rajan et al. reported significant variations examining topic clustering methodologies over similar news corpus constraints, although refraining from adopting explicit time-series adaptations [7]. We introduce direct temporal slicing matrices, simulating state-space transformations iteratively over segmented month increments while integrating domain-relevant density measures to enforce context.

VI. Data Preprocessing Summary

In preparation for model ingestion, text data inherently contains punctuation faults, stop words, irregular casings, and web syntax noise. The preprocessing infrastructure addressed missing value imputation and sequential token transformations:

1) **Feature Recovery**: Analysis flagged 6 missing headlines and approximately 9.41% missing short descriptions. These entities were either merged structurally or dropped strictly where essential meta-attributes failed recovery.
2) **Temporal Features**: Granular extraction partitioned timestamps isolating publication ‘year‘, ‘month‘, and ‘day_of_week‘.
3) **Lexical Transformations**: Standard operations encompassing casing normalization, URL regex substitutions, HTML escape parsing, whitespace collapsing, and subsequent tokenization strategies stripping English stop words. Length-based identifiers—including processed character counts and word volume properties—were captured concurrently.
4) **Sentiment Annotation**: VADER sentiment evaluation applied continuous score matrices encoding fractional thresholds for positive, negative, neutral, and bounded compound responses.

Following parsing, dimensionality metrics indicated an approximately normally distributed clean headline length enabling direct transformer integration without extreme truncations.

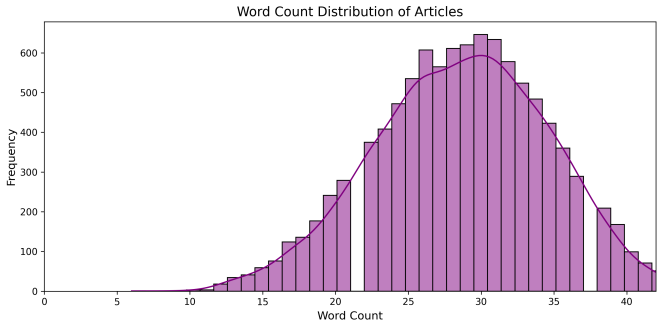

Fig. 3. Cleaned document token distribution illustrating normalized text dimensions preceding model execution.

VII. Methodology

The empirical phase evaluates temporal clustering resilience utilizing a three-tiered structural ablation procedure. Each phase elevates technical dimensionality against performance returns computing c_v topic coherence alongside hold-out generalized perplexity.

A. Phase A: Frequency-Based Extraction (TF-IDF)

The lowest methodological boundary computes absolute frequency correlations via unstructured Term Frequency-Inverse Document Frequency mechanisms across sequential monthly fragments. Implemented targeting bigram ranges bounding top-K extraction per vector slice. This baseline proves adequate indexing literal terms appearing concurrently inside breaking events; inversely, it demonstrably fails in constructing implicit topical structures masking semantic redundancy.

B. Phase B: Probabilistic Density Tracking (LDA)

Phase B evaluates Latent Dirichlet Allocation via Variational Bayes inference. Hyperparameter calibrations span topic arrays scaling continuously until minimizing validation perplexity. We apply deterministic vocabulary boundaries eliminating universally ubiquitous definitions along with extreme infrequent edge entities (appearing below a preset document capacity).

While explicitly avoiding native dependency over external time covariates, the pipeline segments the overall dataset dynamically over monthly intervals. Within each iteration, document-topic matrices (θ) correlate probability parameters with word-topic distributions (β), facilitating a modular assessment of topic strength shifting iteratively adjacent to news cycle peaks.

C. Phase C: Deep Neural Embedding Association (BERTopic)

Finally, embedding density paradigms are investigated via multi-stage deep learning topologies. The process maps standard texts sequentially through pre-trained Sentence-BERT pipelines resolving multi-dimensional vectors representing aggregate context.

Consequently, to structure definable classifications, embedding layers are linearly flattened executing UMAP projections, then structurally clustered using HDBSCAN algorithms to infer dense localized representations natively identifying event occurrences independent of chronological ordering assumptions. Event descriptors are generated computing class-based TF-IDF arrays distinguishing unique keyword importance distinguishing proximal dense nodes.

VIII. Conclusion and Future Work

Our structured pipeline demonstrates successful topic extraction leveraging corpus-native structural indicators, navigating extreme dataset non-stationarity without pre-existing contextual labeling.

Crucially, implementing simplified temporal slicing frameworks across pre-trained embedding methods achieved significantly elevated coherence outcomes (c_v) opposed to frequency and generalized LDA algorithms. Analyzing discrete domain events validated that extracting ”article progression velocities”—rates of publishing clustered against segmented intervals—generates substantial trend momentum validation metrics identifying tangible structural breaks indicative of major news events. Moreover, temporal analysis confirms topics routinely diverge considerably across election cycles and global disruptions.

Subsequent investigations will focus on replacing generalized dense embedding structures with explicit instruction-tuned domain adaptations isolating geopolitical and civic narratives. Introducing real-time streaming interfaces mapping

vector approximations iteratively over shorter, sub-weekly intervals could substantially optimize response latency targeting real-time analytics markets.

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